

9. The Story of Binary Successor

The Problem: Binary Successor

For computer scientists, the computation of binary successor is the basis for nearly all more sophisticated computations.

Intro: Long Addition in the Simplest Case

$$\begin{array}{r} \text{CARRY} \\ 1 \\ + 1 \\ \hline 1 \ 0 \\ \text{SUM} \end{array}$$

A *bit* = a single binary digit (or numeral).

The result of the initial sum is two bits long, which we break into parts.

The *first* bit is called the **SUM** bit (or S bit). The *second* bit is called the **CARRY** bit (or C bit).

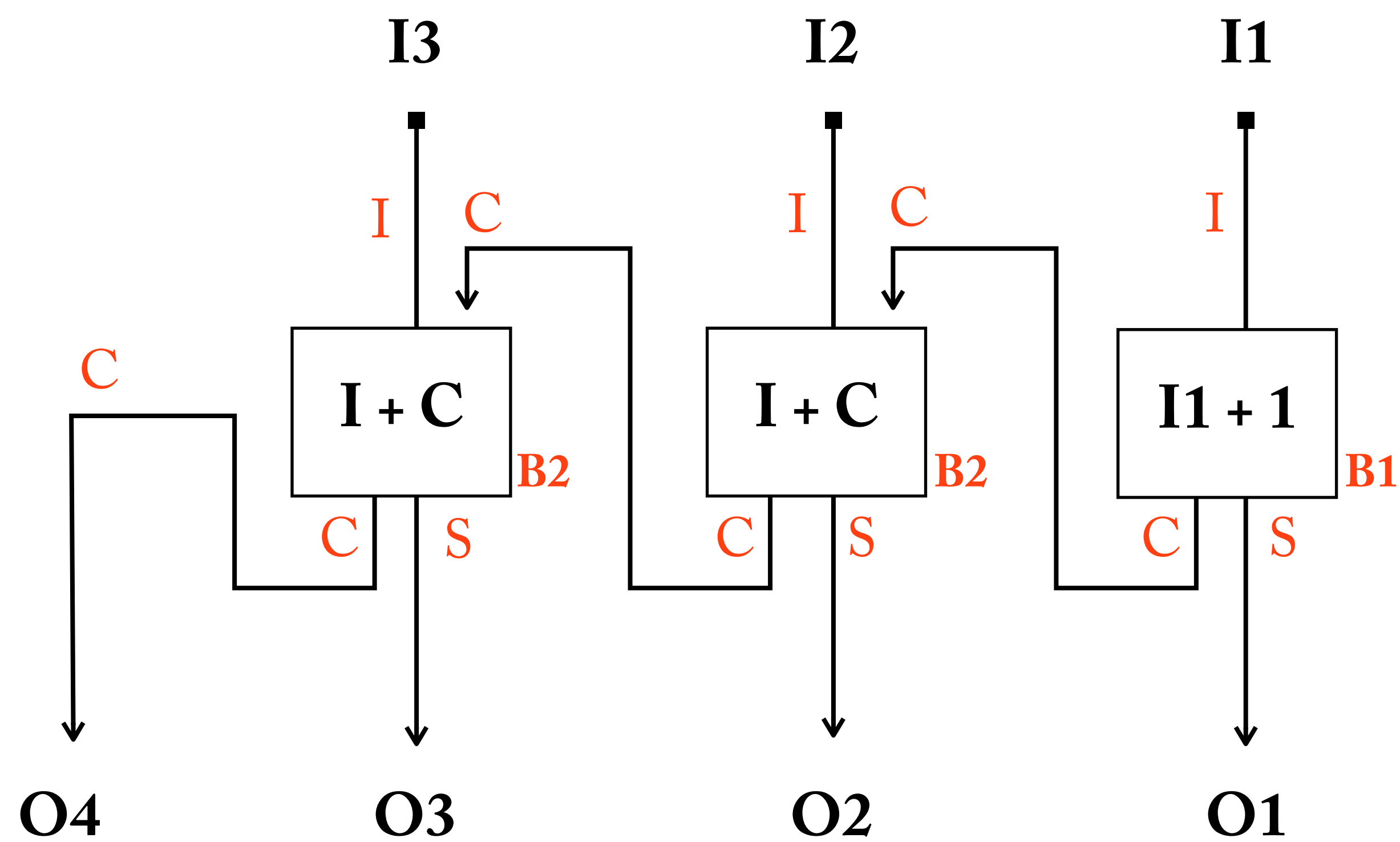
Step 1. Identify a method (carrying)

$$\begin{array}{r} \text{CARRY} \\ \text{C5} \ \text{C4} \ \text{C3} \ \text{C2} \\ 0 \ 0 \ 1 \ 1 \\ \text{I4} \ \text{I3} \ \text{I2} \ \text{I1} \\ 1 \ 0 \ 1 \ 1 \\ + \qquad \qquad 1 \\ \hline 0 \ 1 \ 1 \ 0 \ 0 \\ \text{O5} \ \text{O4} \ \text{O3} \ \text{O2} \ \text{O1} \end{array} \begin{array}{l} \text{INPUT} \\ \text{OUTPUT} \end{array}$$

Binary successor algorithm

1. Compute $I_1 + 1$.
Results in a SUM bit and a CARRY bit.
2. $O_1 = \text{SUM}$.
3. Compute $I_{\text{NEXT}} + \text{CARRY}$.
Results in a new SUM bit and a new CARRY bit.
4. $O_{\text{NEXT}} = \text{SUM}$.
5. Repeat from 3, as long as there are inputs.
6. When there are no inputs left, $O_{\text{FINAL}} = \text{CARRY}$.

Step 2. Break it into parts



We could keep doing this forever...!

Step 3. Analyze the parts

Box 1: $I1 + 1$

Input	Sum	Carry
1	0	1
0	1	0

Box 2: $I + C$

Input	Carry (IN)	Sum	Carry
1	1	0	1
1	0	1	0
0	1	1	0
0	0	0	0

Next you must design a circuit corresponding to each box....

Interlude: Half-Adder

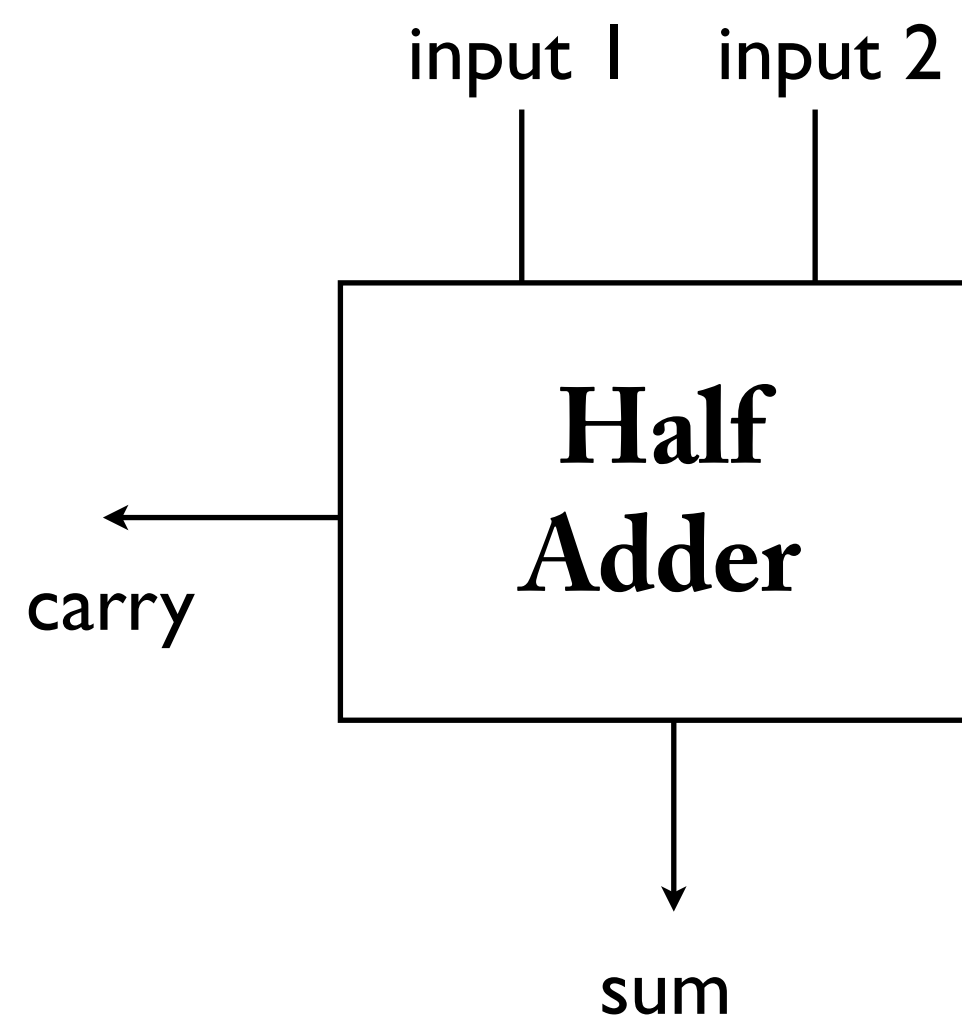
The Box 2 circuit is especially useful and important. It performs a key action: it takes any two bits as inputs, and outputs a SUM bit and a CARRY bit.

This mini-circuit is known as a “half adder”. Half adders form the heart of any addition-based computation.

It is only *half* an adder, because it only adds two bits, rather than two bits *plus* a carry bit.

NOTE: in order to use half adders in your homework solutions, you’ll first have to show how a half adder is built from logic gates.

Schematically, it looks like this:



Always label your SUM and CARRY outputs!

Step 4:A schematic solution

